

Why parliamentary systems collapse under fragmentation: Opposition cohesion, not government formation failure

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Abstract

Parliamentary systems pass more bills than presidential systems at baseline, but collapse to near-zero passage under party-system fragmentation. The literature offers three competing micro-explanations: coalition-formation failure, party discipline, and committee gatekeeping. These operate simultaneously in any real legislature, so observational studies struggle to separate their contributions. We present an agent-based model that compares four democratic institutions: pure parliamentary, pure republican/presidential, premier-presidential (France), and president-parliamentary (Russia). Across four scenarios and $N = 200$ seeds per cell we report bootstrap confidence intervals, Morris screening, Sobol variance decomposition, mechanism ablations, and a hung-parliament variant comparison. Three findings emerge. First, government formation failure alone does not halt legislation: when a fragmented parliament reverts to personal voting, parliamentary passage (46.4%) is statistically indistinguishable from the presidential benchmark (44.8%); collapse requires cohesive opposition obstruction, which drives passage to 0.05%. The fragmentation pathology is therefore a discipline phenomenon operating through the opposition side, not a coalition-formation failure per se. Second, disabling discipline restores fragmented passage to 46.7%, and the rescue magnitude is monotone across the four institutions in a pattern that survives varying the common discipline level. Third, the passage–representation tradeoff is a single spectrum: parliamentary maximises throughput at the cost of representational fidelity; republican maximises fidelity via the presidential veto; semi-presidential variants split the difference.

Keywords: legislative studies, institutional design, party discipline, opposition cohesion, minority government, semi-presidentialism, agent-based modelling, Morris screening, Sobol sensitivity.

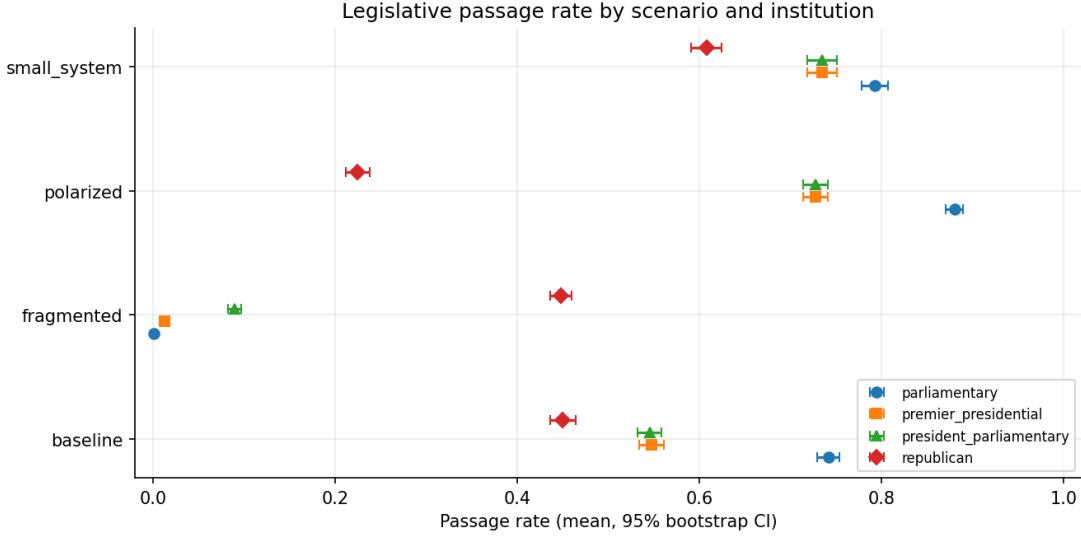


Figure 1: Mean passage rate with 95% bootstrap confidence intervals across four institutions and four scenarios ($N = 200$ seeds per cell). The baseline parliamentary advantage reverses under fragmentation: pure parliamentary and premier-presidential collapse to near-zero passage, while president-parliamentary partially rescues and republican is unaffected. Section 4.1 reports the full passage-rate table; Section 4.2 decomposes the collapse into government-formation failure versus opposition-side cohesion and shows only the latter is causal; Section 4.4 locates the mechanism in party discipline.

1 Introduction

Democracies differ dramatically in how much legislation they produce. In the United Kingdom, over 95% of government bills reach Royal Assent and government defeats on second reading have not occurred since 1986 [12]. Germany passes roughly 90% of government-initiated bills [5]. In contrast, only 3–7% of bills introduced to the U.S. Congress are enacted into law, though most of that figure reflects filtering at committee stage rather than genuine floor rejection [9]. France, a premier-presidential system, enacts roughly 70% of its laws through government-sponsored *projets de loi* [8]. Russia’s State Duma, a president-parliamentary system, passes essentially 100% of presidential bills while opposition initiatives succeed at below 1% [15].

These differences are hard to attribute to demand: mass policy preferences in established democracies, while not identical, overlap heavily and cluster near the centre of the left–right scale [3, 16], yet the legislatures elected by these publics operate at order-of-magnitude different passage rates. The natural explanation is the rules. But the rules interact in ways that are difficult to disentangle. Three mechanistic accounts compete to explain why some legislatures produce laws efficiently and others grind to a halt. The first emphasises **coalition formation**: where a single party cannot hold a working majority, bargaining over portfolios and policy may fail to produce a viable government, blocking any legislative action before it starts [13, 23]. The second emphasises **party discipline**: strong whipping aligns MPs with their leadership rather than their personal or constituent preferences, enabling the government to pass bills that a free vote would reject [1, 4]. The third emphasises **committee gatekeeping**: structural rules that allow small numbers of legislators to kill bills before floor consideration [19, 24]. These accounts are not mutually exclusive, but their relative contributions to any observed passage pattern are difficult to estimate because all three operate simultaneously in every real legislature.

This paper uses an agent-based model (ABM) to attribute passage outcomes to specific mechanisms. An ABM can implement coalition formation, party discipline, and committee gatekeeping

in a single deterministic simulation and disable them one at a time while holding everything else fixed. The mechanism-level counterfactuals required for causal attribution are straightforward in a simulation and impossible in observational data. Prior ABM work in legislative studies has typically modelled a single mechanism at a time: coalition formation [13], electoral mandate structure [7], and spatial voting in the Downs [6] tradition. Our contribution is to integrate the three competing mechanisms in one framework and compare their contributions across institutions and scenarios, building on the stylised pattern that presidential systems achieve lower legislative success rates than prime-ministerial systems while remaining robust to coalition failure [17, 20].

We compare four institutional types. Pure **parliamentary** systems follow the Westminster-Continental European template: the cabinet is drawn from the legislature and falls on failed confidence votes. Pure **republican** (presidential) systems follow the U.S. template: the executive is elected separately, has formal veto power, and is not responsible to the legislature. Between these two sit the **semi-presidential** variants classified by Shugart and Carey [21]. **Premier-presidential** systems (France, Portugal, Ireland) have a directly elected president alongside a cabinet responsible to parliament, but the cabinet is formed by parliamentary majority and the president cannot unilaterally dismiss the prime minister. **President-parliamentary** systems (Russia, Weimar Germany) have the same pair of offices but the president forms the government and can dismiss the PM without a parliamentary vote. In our model both variants are reachable from a single class through two independent configuration toggles.

Across these four institutions we run $N = 200$ random seeds per scenario, report bootstrap 95% confidence intervals, use Morris elementary-effects screening and Sobol variance decomposition to identify the dominant parameters, and run mechanism ablations (disabling committees, discipline, or veto one at a time) to attribute passage contributions. We also check the robustness of the main ordering by varying the common discipline level applied to all institutions.

Three findings structure our contribution. First (§4.2), government formation failure alone does not halt legislation. We compare two defensible readings of a hung parliament: one in which cohesive opposition blocks every bill, and a personal-vote null in which MPs revert to their own preferences when no whip is in play. Under the null, fragmented-parliamentary passage (46.4%) matches the presidential benchmark (44.8%); under obstruction it collapses to 0.05%. The fragmentation pathology is therefore a *discipline* phenomenon operating through the opposition side — not a coalition-formation failure per se. Second (§4.4), the no-discipline ablation restores parliamentary passage from 0.05% to 46.7%, and the rescue magnitude is monotone across the four institutions in the order parliamentary > premier-presidential > president-parliamentary > republican, an ordering that holds at all tested common discipline levels and reverses sign under polarisation: disabling discipline there costs parliamentary 66 percentage points. Discipline aggregates votes wherever a coalition exists; its absence or presence on the *opposition* side decides whether a hung parliament legislates at all. Third (§4.5), the passage–representation tradeoff is a single spectrum. Parliamentary passes the most bills but its passed-bill ideology matches no particular filter (policy representation gap $\approx +0.04$ at polarised). Republican filters aggressively through the presidential veto (gap ≈ -0.53 at polarised) at the cost of high non-passage rates. Semi-presidential variants sit between.

The remainder of the paper proceeds as follows. Section 2 describes the model; Section 3 describes the experimental design; Section 4 presents the results; Section 5 discusses implications and relates the findings to prior work; Section 6 states limitations; Section 7 documents code, data, and the interactive companion app.

2 Model description

2.1 Agents and state

The model contains four agent types: **constituencies** with a fixed two-dimensional ideology (economic $\times$ social) and a population; **parties** with an ideology and a name; **legislators** with an ideology, an assigned constituency, and an assigned party; and **committees** with a policy-area jurisdiction (an ideology centre and radius) and party-proportional membership. Bills are ephemeral objects with an ideology and a salience, instantiated per legislative step.

All four institutions share the same agent types and initialisation procedure. They differ only in the bill-processing pipeline invoked by `pass_legislation(bill)` and in the institution-specific state that pipeline reads and writes (government, cabinet, president, confidence votes, vetoes, dismissals).

Initialisation is deterministic. Constituencies, parties, and legislators each receive evenly spaced ideologies along the main diagonal of the space, from $(-1, -1)$ to $(1, 1)$; legislator i is assigned to party $i \bmod n_{\text{parties}}$ and constituency $i \bmod n_{\text{constituencies}}$. The round-robin party assignment means every party’s members span the full ideological range: parties are maximally heterogeneous internally, so the whip has real aggregation work to do. There are no stochastic initial conditions; the seed drives only process randomness (bill draws, vote rolls, committee sampling). Section 4.6 reports a robustness check that replaces this spread initialisation with party-clustered legislator ideologies.

2.2 Passage pipelines

Parliamentary: bills route to a committee with matching jurisdiction (committees can approve, amend, or kill); with probability `confidence_matter_rate` a bill is treated as a confidence vote and whipped harder; a floor vote with party discipline decides passage; failed confidence votes trigger government collapse and reformation.

When no governing coalition can be formed at all — the fragmented scenario makes this certain for majority-driven institutions — the whip structure is undefined, and two readings of the situation are possible:

- **cohesive obstruction**: every MP counts as “opposition” and votes against bills with probability `discipline  $\times$  multiplier`, an anti-system reading in which polarised blocs block all legislation [2, 18];
- **personal vote**: no whip is in play, so every MP reverts to their own preference, the issue-by-issue-majority world of minority-government governance [13, 22, 23].

A configuration flag, `hung_parliament_behavior` $\in \{\text{cohesive_obstruction}, \text{personal_vote}\}$, selects between them; the shipped default is cohesive obstruction. Section 4.2 shows that this single choice determines whether fragmentation produces a collapse at all, which is why we treat it as an experimental factor rather than a silent implementation detail.

Republican: bills route to a committee (with stronger gatekeeping power); a floor vote with weak discipline decides legislative passage; the executive vetoes probabilistically by ideological distance from the bill; a two-thirds override attempt follows vetoes.

Semi-presidential: hybridises both pipelines. Committees route bills; bills may be treated as confidence matters; a floor vote with moderate discipline decides legislative passage; the directly-elected president then checks a veto; if the president can dismiss the PM (president-parliamentary

variant) the dismissal check fires on every bill and may collapse the government independently of confidence votes.

2.3 Configuration toggles for semi-presidential variants

Two `SemiPresidentialConfig` toggles distinguish the variants:

- `government_formation` $\in \{\text{majority_driven}, \text{president_driven}\}$. Majority-driven formation follows the parliamentary rule (largest party alone if it holds a majority, else two-party coalition). President-driven formation appoints the president’s party as the governing party, adding a second party for majority only if available; otherwise a minority cabinet.
- `president_can_dismiss_pm` $\in \{\text{False}, \text{True}\}$. When `True`, the president dismisses the PM with probability `presidential_dismissal_rate` whenever their ideological distance exceeds 0.5.

Premier-presidential corresponds to `(majority_driven, False)`; president-parliamentary to `(president_driven, True)`. Non-canonical combinations are reachable but not the focus here.

2.4 ODD protocol

A full ODD description [10, 11] appears in Appendix A. All agent states, scheduling, design concepts, initialisation, and submodel equations are documented there.

3 Experimental design

3.1 Scenarios

Four scenarios vary the political environment while holding institutional rules fixed: **baseline** (20 legislators, 3 parties, 6 constituencies, 25 bills, bill ideology range $[-1, 1]$); **fragmented** (24 legislators, 5 parties, 8 constituencies, 30 bills, range $[-1, 1]$); **polarised** (16 legislators, 2 parties, 4 constituencies, 20 bills, range $[-1.5, 1.5]$); and **small-system** (10 legislators, 2 parties, 3 constituencies, 15 bills, range $[-0.8, 0.8]$).

3.2 Statistical harness

Each (scenario, institution) cell is replicated over $N = 200$ seeds. We report means with percentile bootstrap 95% confidence intervals (`scipy.stats.bootstrap`, 9999 resamples). Pairwise institution comparisons use Welch’s t -test and Mann-Whitney U ; effect sizes are reported as Cohen’s d . When zero-variance cells produce undefined p -values, those are recorded as `NaN` rather than imputed.

3.3 Sensitivity analysis

For each institution we define a six-parameter problem and run Morris elementary-effects screening ($r = 20$ trajectories, 4 levels) and Sobol variance decomposition ($N = 256$ base samples, first-order and total-order indices). Three seeds per design point reduce stochastic variance without inflating cost. The six parameters per institution always include `discipline_strength`, `committee_gatekeeping_power`, `num_parties`, and `num_constituencies`; the remaining two are institution-specific.

3.4 Mechanism ablations

Three ablations modify the running model without touching the production code. `no_committees` replaces the committee-routing function with a pass-through; `no_discipline` sets `discipline_strength = 0`; `no_veto` (where applicable) replaces the executive-veto check with `False`. Each ablation is monkey-patched on the live model instance; no class-level mutation occurs.

3.5 Discipline-default robustness check

To test whether the observed ordering depends on our default discipline values, we sweep a common discipline level $D \in \{0.0, 0.1, \dots, 0.9\}$ applied simultaneously to all four institutions, and compute $\text{rescue}(D) = \text{passage}(D = 0) - \text{passage}(D)$ per institution at $N = 100$ seeds per cell.

3.6 Hung-parliament variant comparison

The fragmentation scenario makes coalition formation impossible for the majority-driven institutions (largest party $\approx 21\%$ of seats; two-party combinations also fall short), so the whip structure defined in §2 never engages. We run the full 4×4 grid at $N = 200$ seeds under both values of `hung_parliament_behavior`. Because the flag can only bind when the coalition list is empty, the two variants are bit-identical in the baseline, polarised, and small-system scenarios (where governments form) and for republican (no formation gate) and president-parliamentary (the president always seats at least a minority cabinet); all differences are therefore concentrated in the fragmented cells for parliamentary and premier-presidential. The runner uses the same bill-generation order as the main harness, so default-variant rows reproduce the main tables exactly.

4 Results

4.1 Passage rates across scenarios

Figure 1 (placed on the title page) summarises per-scenario mean passage rates with 95% bootstrap CIs for all four institutions. At baseline the expected ordering holds: parliamentary 0.742 [0.729, 0.754], premier-presidential 0.547, president-parliamentary 0.546, republican 0.450 [0.436, 0.464]. Parliamentary’s advantage (+29pp) over republican is highly significant (Welch’s $t = 38.9$, $p < 10^{-200}$; Cohen’s $d = 2.88$).

4.2 The fragmentation collapse decomposed

Under the shipped default (cohesive obstruction), fragmentation collapses majority-driven institutions: parliamentary passage falls to 0.0005 (95% CI [0, 0.0012]), premier-presidential to 0.013, while president-parliamentary partially rescues passage to 0.086 through its president-driven minority-cabinet rule and republican holds at 0.448 (Figure 3; the collapse is a regime property, not a tail event).

But which ingredient of fragmentation does the collapsing? Formation failure and whip behaviour are confounded in the default runs, because both are absent simultaneously. Varying `hung_parliament_behavior` unconfounds them (§3.6). Table 1 and Figure 2 show the result: under the personal-vote null, fragmented parliamentary passage *recovers* to 46.4% [45.3, 47.6] — statistically indistinguishable from the presidential benchmark of 44.8% [43.6, 46.0] — while under cohesive obstruction it stays collapsed at 0.05%. Premier-presidential moves from 1.3% to 33.3%

under the null, below parliamentary because the presidential veto filters bills even without obstruction. Republican and president-parliamentary are identical across variants, as required: neither can enter a hung state.

Table 1: Passage rate under fragmentation by hung-parliament behaviour ($N = 200$ seeds per cell). The flag binds only where the coalition list can be empty.

Institution	Cohesive obstruction	Personal vote	Δ
parliamentary	0.0005	0.464	+0.464
premier_presidential	0.013	0.333	+0.320
president_parliamentary	0.086	0.086	0
republican	0.448	0.448	0

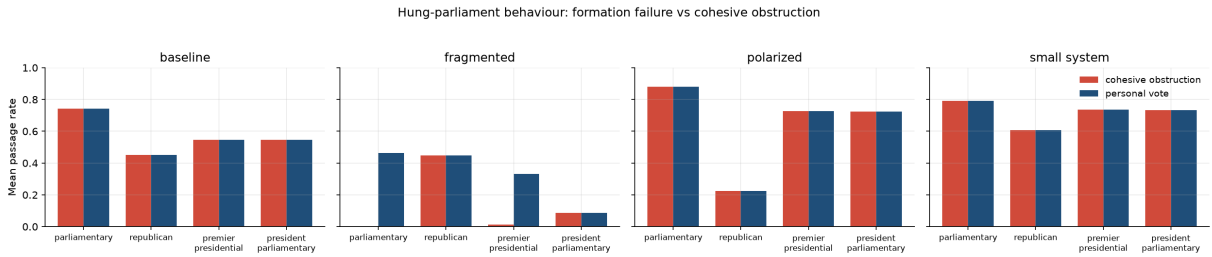


Figure 2: Mean passage rate by institution and hung-parliament behaviour, per scenario ($N = 200$ seeds). Variants differ only in the fragmented cells of the majority-driven institutions: formation failure alone leaves parliamentary passage at presidential levels (46.4% vs. republican’s 44.8%); cohesive opposition obstruction produces the collapse.

The decomposition reverses the causal story told by observational comparisons. Coalition-formation failure, emphasised in the government-formation literature [13, 23], is *not sufficient*: a fragmented chamber whose members vote their preferences legislates at full presidential-system rates. What generates the near-zero regime is cohesive obstruction on the opposition side — the anti-system pattern in which blocs whip against all business regardless of content [2, 18]. This matches the empirical record: fragmented parliaments with weakly-disciplined or office-seeking oppositions keep passing legislation through minority and caretaker cabinets, while polarised anti-system blocs produce legislative deadlock.

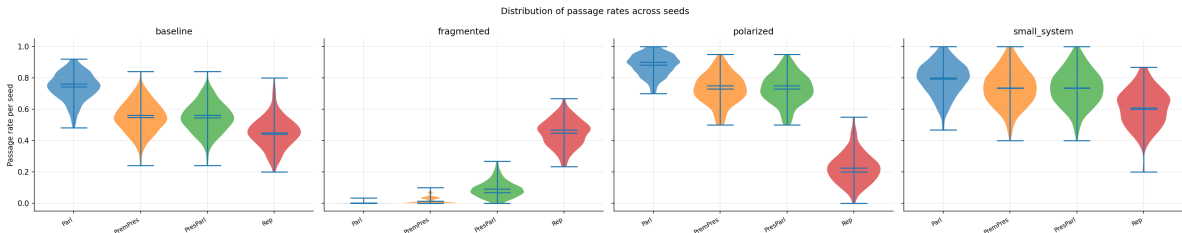


Figure 3: Per-seed passage-rate distributions, faceted by scenario. The fragmented cell shows near-zero passage for both majority-driven institutions (parliamentary, premier-presidential) and a partial rescue under president-parliamentary.

4.3 Sensitivity: different institutions, different bottlenecks

Sobol total-order indices at baseline (Fig. 4) show that parliamentary passage is dominated by `num_parties` ($S_T = 0.808$), consistent with the fragmentation-collapse finding. Republican passage is dominated by `committee_gatekeeping_power` ($S_T = 0.671$) with `discipline_strength` second ($S_T = 0.477$). The two institutions have structurally different bottlenecks. Semi-presidential variants track parliamentary in their dependence on `num_parties`.

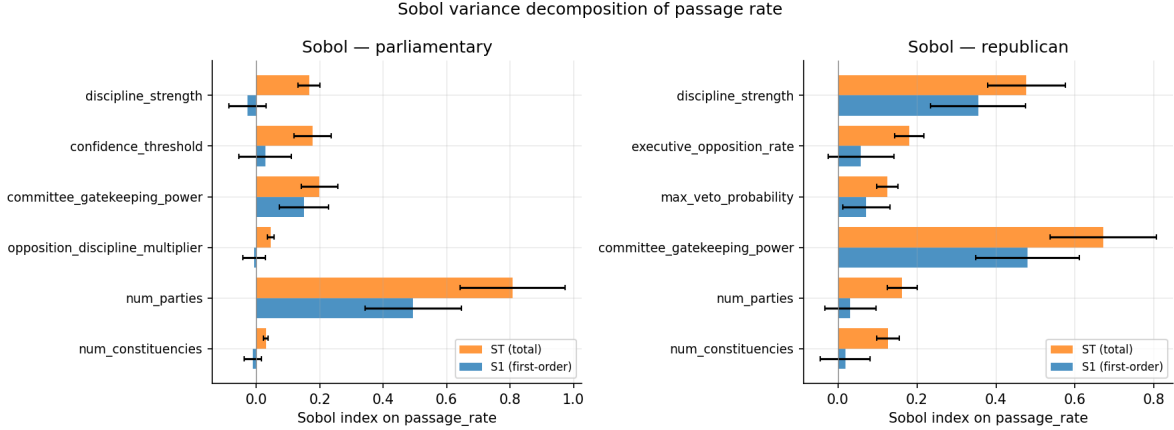


Figure 4: Sobol first-order (S_1) and total-order (S_T) indices for passage rate at baseline.

The same analysis on the fragmented scenario confirms the `num-parties` dominance across all three majority-dependent institutions ($S_T \approx 0.76$ – 0.79). `confidence_threshold` gains importance under fragmentation (+0.12 in S_T for parliamentary), which is consistent with a binding majority-threshold interpretation. Republican remains structurally insensitive to `num_parties` ($S_T = 0.14$).

4.4 Mechanism attribution via ablation

Figure 5 shows $\Delta \text{passage_rate}$ relative to the full model for each (scenario, institution, ablation) cell at $N = 200$. The headline result appears at fragmented:

Table 2: `no_discipline` ablation Δ under fragmentation.

Institution	Full-model passage	Δ (no_discipline – full)
parliamentary	0.0005	+0.466
premier_presidential	0.013	+0.312
president_parliamentary	0.086	+0.233
republican	0.448	−0.242

Disabling party discipline restores parliamentary passage from near-zero to nearly 47% — closely matching the personal-vote hung parliament of §4.2 (46.4%) via two independent manipulations (zeroing the whip everywhere vs. removing it only where no government exists), which cross-validates the attribution. The rescue magnitude decreases monotonically across the four institutions in the order parliamentary > premier-presidential > president-parliamentary > republican. Republican is the only institution where zeroing discipline *hurts* rather than helps: without a majority-formation gate, discipline plays a different role (crude vote aggregation on the floor) and removing it simply damages the signal.

Under polarised scenarios the same ordering holds in magnitude but with opposite sign (Table 3). Discipline *enables* passage when the coalition wants extreme bills to pass; disabling discipline costs parliamentary 66 percentage points. Republican, with no coalition to whip, loses only 11 percentage points. Across scenarios, discipline is the mechanism majority-dependent institutions use to *aggregate votes*, rather than a mechanism that specifically causes fragmentation pathologies.

Table 3: no_discipline ablation Δ under polarisation.

Institution	Full-model passage	Δ (no_discipline – full)
parliamentary	0.880	−0.664
premier_presidential	0.728	−0.557
president_parliamentary	0.723	−0.557
republican	0.225	−0.113

4.5 Passage–representation tradeoff

Figure 6 plots passage rate against a new metric, the policy representation gap:

$$\text{gap} = \overline{d_{\text{passed}}} - \overline{d_{\text{proposed}}} \quad (1)$$

where d is the L2 distance between a bill’s ideology and the constituency-median ideology. Negative gaps indicate the institution filters passed bills toward the median; positive gaps indicate the institution spreads passed bills relative to random proposal draws.

The tradeoff is sharpest under polarised scenarios:

Table 4: Passage–representation tradeoff at polarised ($N = 200$).

Institution	Passage rate	Representation gap
parliamentary	0.882	+0.037
premier_presidential	0.725	−0.012
president_parliamentary	0.730	−0.010
republican	0.218	−0.530

Parliamentary passes 88% of bills with essentially no ideological filter; republican passes only 22% but the bills that do pass are strongly biased toward the constituent median, reflecting the presidential veto’s role as an ideological screen. Semi-presidential variants sit on the same spectrum between them. Filtering and throughput are mechanistically linked: any mechanism that increases one reduces the other.

4.6 Robustness of the monotone ordering

The ablation ordering in Table 2 could in principle be an artefact of our default `discipline_strength` values, which happen to be ordered the same way across institutions (0.8, 0.6, 0.6, 0.4). Figure 7 tests this by setting all four institutions’ `discipline_strength` to the same value $D \in \{0.1, \dots, 0.9\}$ and measuring the rescue magnitude.

The monotone ordering parliamentary > premier-presidential > president-parliamentary > republican holds at 9/9 tested D values under fragmentation. The structural claim survives: the

ordering is not a function of our chosen defaults. A regression test in the repository enforces this at $D \in \{0.0, 0.2, 0.5, 0.8\}$ so that future mechanism changes that break the ordering fail CI.

A second initialisation concern is the spread assignment itself (§2): evenly spaced legislators with round-robin party membership make parties maximally heterogeneous internally, which plausibly inflates the *magnitude* of the discipline effect — the whip has more work to do when members span the whole spectrum. We therefore re-initialise legislator ideologies as their own party’s position plus Gaussian noise ($\sigma = 0.15$) and re-run the discipline ablation ($N = 100$ seeds per cell). Both findings keep their structure. Under fragmentation the rescue ordering survives: parliamentary +0.35, premier-presidential +0.23, president-parliamentary +0.16, republican ≈ 0 . Under polarisation the sign flip survives: parliamentary -0.74 , semi-presidential variants near -0.57 , republican flat. Magnitudes attenuate relative to the spread initialisation (parliamentary’s fragmented rescue falls from +0.47 to +0.35), exactly as the heterogeneity account predicts. Within-party heterogeneity scales the size of the discipline effect but not its ordering or its sign structure.

5 Discussion

5.1 Discipline as a universal aggregation mechanism

The hung-parliament decomposition (§4.2) and the ablation results across fragmented and polarised scenarios (Tables 2 and 3) reveal that party discipline is not simply a “collapse trigger” under fragmentation or a “passage enabler” under polarisation. It is the same mechanism in both cases: discipline aggregates legislators toward a bloc position rather than the free-vote median. When that bloc is a governing coalition near the floor median, discipline enforces passage. When it is an opposition bloc facing a chamber with no government, the same aggregation produces uniform obstruction. And when no bloc exists at all — the personal-vote null — a fragmented chamber legislates at full presidential rates, so formation failure alone blocks nothing.

The monotone ablation ordering across the four institutions reflects how much each relies on a formation-gated coalition to act. Parliamentary requires a majority coalition before any legislation is possible; premier-presidential requires the same, but with a presidential veto adding a second gate; president-parliamentary also nominally requires a majority, but the president-driven formation rule lets the government function as a minority cabinet when no majority forms; republican has no formation gate at all. The rescue magnitude under `no_discipline` scales with this dependence. This reading is consistent with Strøm [22, 23]’s demonstration that minority governments are rational, functioning cabinet solutions rather than instability artifacts — our personal-vote variant is a stylised version of their issue-by-issue-majority world — and with Shugart and Carey [21]’s typology of semi-presidential variants.

The obstruction side of the decomposition has older roots. Sartori [18]’s anti-system parties — blocs defined by opposition to the democratic rules of the game rather than by policy competition — generate exactly the uniform against-voting our cohesive-obstruction variant encodes, and Capoccia [2] documents how interwar extremist parties used parliamentary presence to deadlock the institutions they opposed. The contrast between the two variants thus maps onto a substantive empirical question about any given fragmented parliament: is its opposition composed of office-seeking parties that bargain issue-by-issue, or of anti-system blocs that obstruct? Our model predicts opposite passage regimes for the two.

This framing also makes a prediction that a pure “fragmentation-collapse” framing does not: under polarisation, the same institutions that suffered the fragmentation collapse should benefit most from discipline. Table 3 confirms this.

5.2 The passage–representation tradeoff as a single spectrum

Figure 6 and Table 4 make a concrete claim about the tradeoff between legislative throughput and representational fidelity: the two quantities lie on a single axis for the four institutions studied. Parliamentary sits at one end (high throughput, low filter); republican at the other (low throughput, high filter); semi-presidential variants between them. The mechanism is straightforward. Discipline-plus-majority passage, the parliamentary mechanism, has no ideological test: whatever the coalition proposes passes. The presidential veto, the republican mechanism, *is* an ideological test: bills far from the executive’s ideology are filtered probabilistically. Committees add filtering in both but the effect is smaller than the presidential veto.

This casts the parliamentary “advantage” in baseline passage rates differently. Parliamentary systems pass more bills in absolute terms but do not necessarily pass *better* bills in the sense of matching constituent preferences. The comparison requires joint consideration of both axes, not rank-ordering on one.

5.3 Policy and institutional-design implications

The conventional response to fragmented-parliament gridlock is *electoral reform*: raise thresholds, move toward first-past-the-post, encourage larger parties. Our results suggest a second route: *discipline reform*. Allowing free votes on more bills, weakening leadership whip powers, or restructuring MP incentives toward their districts rather than their parties could restore passage capacity under fragmentation without touching electoral rules. Whether such reforms also damage representation is an open empirical question: the parliamentary filter is already weak in our model, so weakening discipline further might spread passed bills without the presidential-veto compensation. This model can illuminate that question but cannot yet answer it directly.

For new constitutional design, the Phase F robustness result suggests a trade-off between fragility and filtering. Pure parliamentary maximises passage at baseline but has a failure mode under fragmentation; president-parliamentary partially insulates against that failure mode but sacrifices some passage at baseline; republican is robust at all scenarios but pays a throughput tax. Premier-presidential is closer to parliamentary than its name suggests and shares the fragmentation vulnerability.

5.4 Relation to prior work

Linz [14] argued parliamentary systems outperform presidential systems in producing governments and policy. Our baseline results (0.742 vs. 0.450) echo that finding on passage rates, but our fragmentation results reverse it: parliamentary collapses while presidential holds steady. The conditional reversal recovers a pattern Linz’s categorical claim cannot predict. Shugart and Carey [21] argued semi-presidential variants should differ from each other in specific ways; we confirm that president-parliamentary’s partial rescue of passage under fragmentation is a direct consequence of the president-driven formation rule. Tsebelis [24] argued additional veto players reduce policy mobility; our tradeoff finding refines this by showing that veto players also improve representation quality, so the choice is not “veto players good or bad” but “veto players trade throughput for filtering.”

6 Limitations

The model is a stochastic pipeline model of institutional rules, not a dynamic population model: legislators, parties, and constituencies are static after initialisation, there are no elections, and no

agent learns or adapts. All institutional differences therefore reflect rule-level filtering of an exogenous bill stream, and nothing here speaks to how representation evolves under electoral feedback; adding endogenous selection (electoral deselection of mavericks, party splits, constituency updating) is the natural next step and would make discipline strength an outcome rather than a parameter.

The hung-parliament behaviour flag resolves a genuine ambiguity rather than a measured parameter: real fragmented chambers mix obstruction and issue-by-issue bargaining, and the two pure regimes we compare are bounds, not descriptions of any single chamber. The shipped default (cohesive obstruction) reproduces the collapse regime reported in the comparative literature, but the personal-vote null is the more conservative reading and, in our view, the better baseline for future work.

The model treats legislator ideology as static and does not include learning or adaptation. It has no endogenous party emergence, realignment, or exit; the party structure is fixed at initialisation. It has no bicameralism, judicial review, federalism, or presidential agenda-setting powers beyond the veto. Bills have no path dependence: each bill is an independent draw. We do not model pre-parliamentary agenda control (e.g., cabinet screening of government bills before introduction), which is why our parliamentary baseline rate of 74% is below the empirical UK and German benchmarks of 90–95% for government bills.

Cohabitation is detected in our semi-presidential runs (shares of 12–26% across scenarios) but does not produce meaningful passage effects: our dismissal rate of 5% per bill under presidential-parliamentary is too gentle to create sustained gridlock, and the premier-presidential presidential veto is too soft to differentially block bills under cohabitation. Future work should treat cohabitation dynamics more carefully, including cabinet-to-legislator ideological alignment and veto thresholds that shift under divided government.

The sensitivity analysis was run primarily on the baseline scenario; a secondary run on the fragmented scenario (§4.3) confirms the `num_parties` dominance but is not a substitute for per-scenario sweeps of all parameters. The Morris and Sobol samplers include `num_parties` and `num_constituencies` in the sweep space; scenario definitions are therefore overlapping with the sensitivity space, which is a methodological choice to allow continuous interpolation rather than a discrete scenario lookup.

7 Reproducibility

All results are deterministic under seed. The repository <https://github.com/tofuadmiral/institutional-representation-abm> contains the full code, 68 automated tests (Python 3.13 on Ubuntu, GitHub Actions CI), a pinned regression fixture, a Streamlit interactive companion application, and a CoMSES metadata card. The command

```
python -m experiments.multiseed_comparison --scenarios all --seeds 200
```

reproduces Figure 1. Table 1 and Figure 2 are produced by `experiments.hung_parliament`; Tables 2 and 3 require additional invocations of `experiments.ablation`; Figure 7 requires `experiments.discipline_robustness`; and Figure 6 requires `experiments.representation`. All entry points are documented in the repository `README.md`. Full runtime is under five minutes on eight cores. An interactive Streamlit companion (`streamlit run streamlit_app/app.py`) exposes every tunable configuration parameter as a slider and reproduces the scenario-comparison, parameter-sweep, and ablation workflows used in §4.

A ODD Protocol

The full ODD protocol [10, 11] is available in the repository at `docs/ODD_PROTOCOL.md`. Core elements are summarised in §2.

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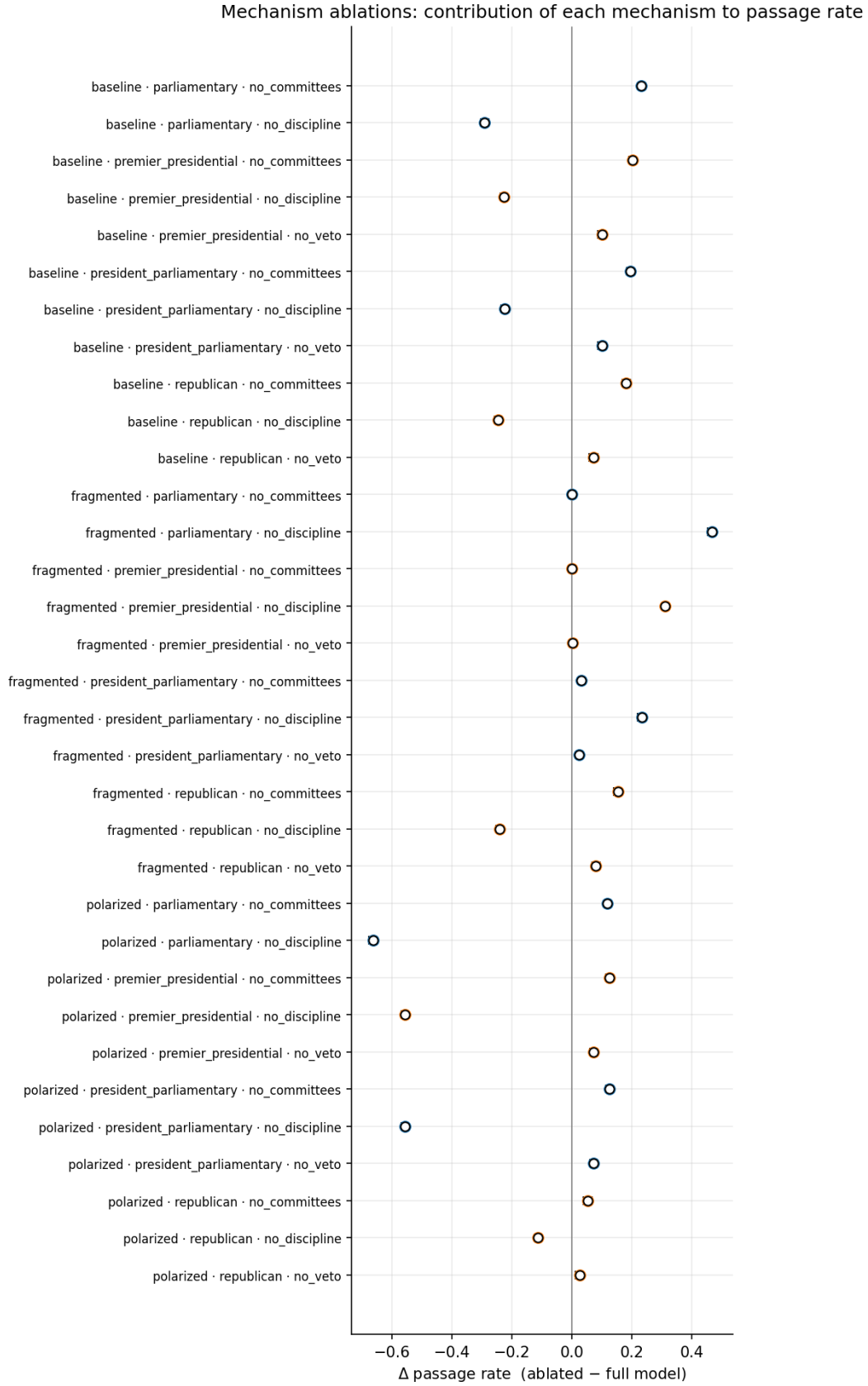


Figure 5: Ablation deltas (Δ passage rate relative to full model) for each (scenario, institution, ablation) cell at $N = 200$ seeds. Confidence bars are ± 1.96 SE.

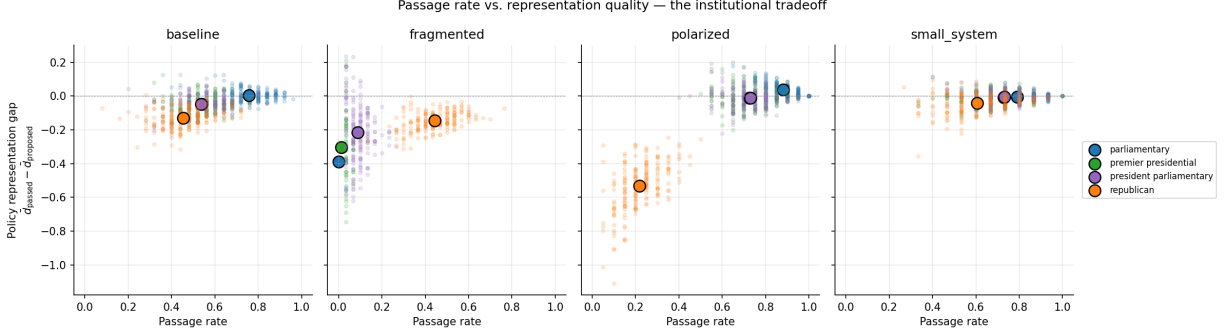


Figure 6: Passage rate versus policy representation gap, per scenario. Each large marker is the $N = 200$ mean; the cloud is per-seed. Under polarisation the tradeoff is most visible: parliamentary at $(0.88, +0.04)$ vs. republican at $(0.22, -0.53)$.

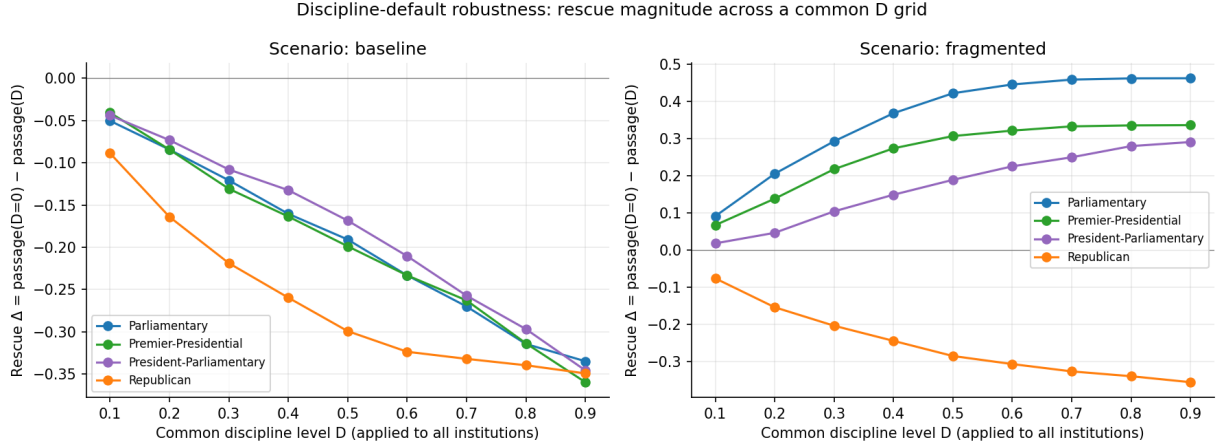


Figure 7: Rescue magnitude versus common discipline level D , per institution, at baseline and fragmented scenarios. Under fragmented (right panel), the monotone ordering parliamentary $>$ premier-presidential $>$ president-parliamentary $>$ republican holds at all nine tested D values.